



Semester 2 Examinations 2015/2016

Exam Code(s) 1MEC1, 1HDS1, 3BMS2, 3BS9, 4BME1, 4FM2
Exam(s) 3rd Science, 4th Science, 4th Financial Mathematics and Economics, 4th Mathematics and Education,

Module Code(s) MP491
Module(s) Nonlinear Systems

Paper No. 1
Repeat Paper No

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Internal Examiner(s) Dr P. Piironen

Instructions: Marks are awarded for Question 1 in Section A **and** the TWO best questions in Section B.
All questions carry the same weight.

Duration 2 hours
No. of Pages 5
Discipline(s) Applied Mathematics
Course Co-ordinator(s)

Requirements:

Release in Exam Venue Yes ☒ No ☐

MCQ Yes ☐ No ☒

Handout None
Statistical/ Log Tables Yes
Log Graph Paper None
Other Materials Calculator

Section A - Answer all parts (a)-(d). (5 Marks/part)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = x^3 + 3x^2 + 2x.$$

Find all equilibrium points and determine their stability.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 5 & -2 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point $(x, y) = (0, 0)$ and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations

$$\begin{aligned} \frac{dx}{dt} &= y^2 - x - 1, \\ \frac{dy}{dt} &= 1 - y^2 - x. \end{aligned}$$

Find and classify the equilibrium points.

- (d) Consider the map

$$x_{n+1} = x_n(x_n - 1)(x_n + 1)$$

for $n = 0, 1, 2, \dots$. Find all fixed points and determine if they are stable or unstable.

Section B - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = x(x - a(a^2 - 1)),$$

where a is a real parameter.

- i. Find all equilibrium points.
- ii. Sketch the bifurcation diagram, and indicate the behaviour of non-equilibrium solutions using appropriate arrows.
- iii. Find all bifurcation points and classify them.

10 Marks

- (b) Consider the system of ordinary differential equations

$$\begin{aligned}\dot{x} &= 2ax - 2x + y, \\ \dot{y} &= -x - y + ay - x^2y,\end{aligned}$$

where $a \in \mathbb{R}$.

- i. Find the values of $a \in \mathbb{R}$ for which the origin $(x, y) = (0, 0)$ is a node.
- ii. Find the value of $a \in \mathbb{R}$ for which a Hopf bifurcation occurs at the origin $(x, y) = (0, 0)$.
- iii. Describe how the nature of the equilibrium point changes at the Hopf bifurcation point found in part ii for varying a .

10 Marks

3. Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= 2xy, \\ \frac{dy}{dt} &= x^2 + y^2 - 1.\end{aligned}$$

(a) Locate and classify all equilibrium points in the associated phase plane.

4 Marks

(b) Identify, and sketch separately, the isoclines corresponding to $k = -1, 0, 1, \infty$, where

$$\frac{dy}{dx} = k.$$

12 Marks

(c) By using all four isoclines found in (b), sketch a sufficient number of phase plane curves to illustrate the phase-plane portrait for this system.

4 Marks

4. Consider the logistic map

$$x_{n+1} = ax_n(2 - x_n), \quad n = 0, 1, 2, \dots,$$

where $0 \leq a \leq 2$ is a real positive parameter and $0 \leq x_0 \leq 2$.

- (a) Investigate for what values of a the logistic map has **one** and **two** fixed points and investigate their stability.

6 Marks

- (b) Determine for what values of a the logistic map has period-2 points and what those points are. For what values of a are the resulting period-2 orbits stable and unstable? Discuss what happens as a increases through the critical value at which the period-2 orbits become unstable?

10 Marks

- (c) Describe what is meant by a *period-doubling cascade* in relation to the logistic map. Sketch, reasonably carefully, the bifurcation diagram you get for the logistic map when varying a between 0 and 2.

4 Marks